

Lab 2 - Task 6

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

aqi = pd.read_csv("city_day.csv")
crop = pd.read_csv("crop_production.csv")

aqi["Date"] = pd.to_datetime(aqi["Date"])

aqi["Year"] = aqi["Date"].dt.year

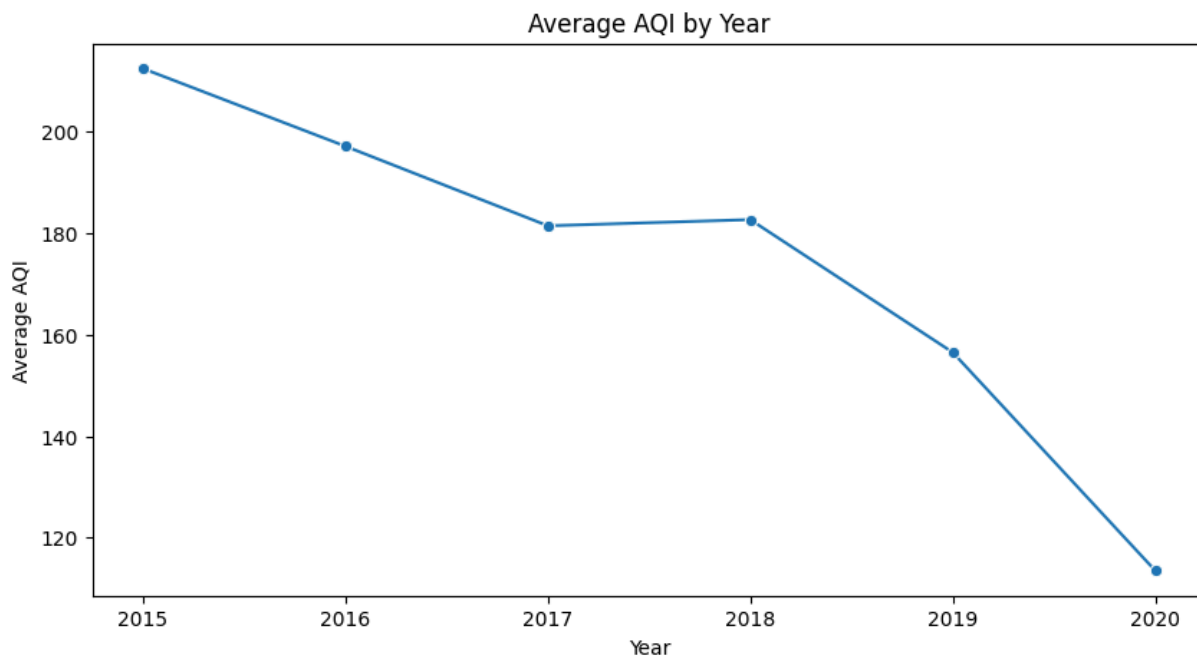
yearly_aqi = (
    aqi.groupby("Year")["AQI"]
    .mean()
    .reset_index()
)

plt.figure(figsize=(10,5))

sns.lineplot(
    data=yearly_aqi,
    x="Year",
    y="AQI",
    marker="o"
)

plt.title("Average AQI by Year")
plt.ylabel("Average AQI")
plt.show()

print(yearly_aqi)
```



	Year	AQI
0	2015	212.463054
1	2016	197.150019
2	2017	181.472789
3	2018	182.684312
4	2019	156.518173
5	2020	113.520697

Lab 2 - Task 7

```
In [3]: aqi["Month"] = aqi["Date"].dt.month

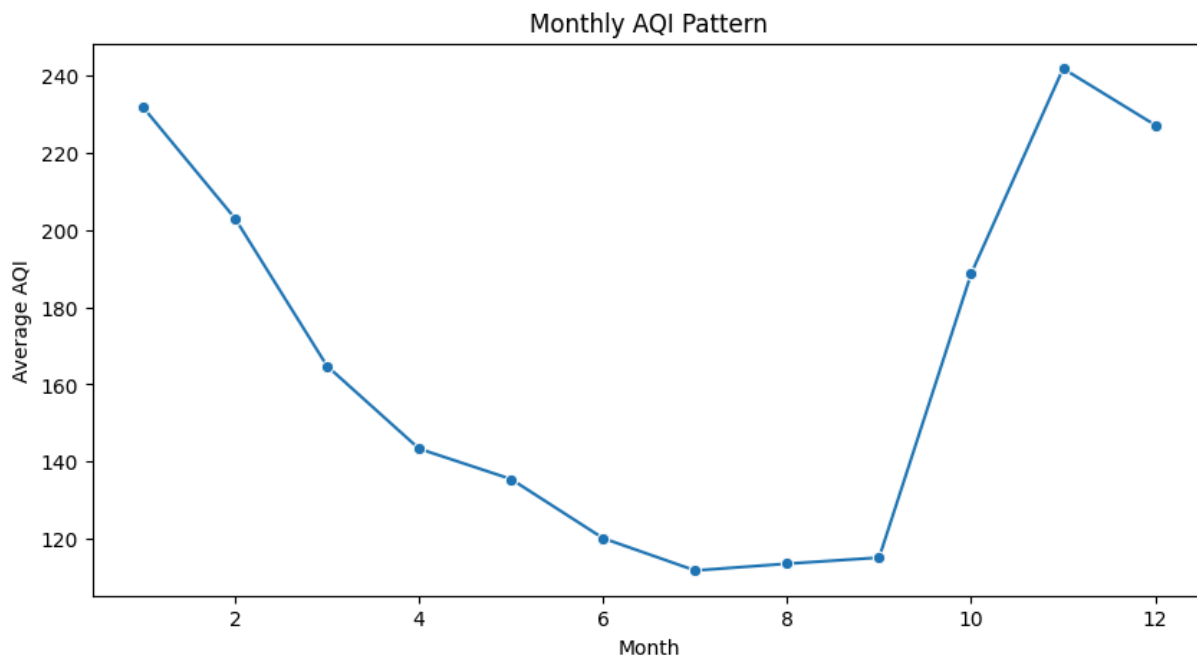
monthly_aqi = (
    aqi.groupby("Month")["AQI"]
    .mean()
    .reset_index()
)

plt.figure(figsize=(10,5))

sns.lineplot(
    data=monthly_aqi,
    x="Month",
    y="AQI",
    marker="o"
)

plt.title("Monthly AQI Pattern")
plt.ylabel("Average AQI")
plt.show()

print(monthly_aqi)
```



	Month	AQI
0	1	231.674918
1	2	202.905197
2	3	164.735281
3	4	143.355120
4	5	135.489579
5	6	120.198379
6	7	111.854575
7	8	113.613176
8	9	115.191804
9	10	188.613552
10	11	241.681302
11	12	227.084980

Lab 2 - Task 8

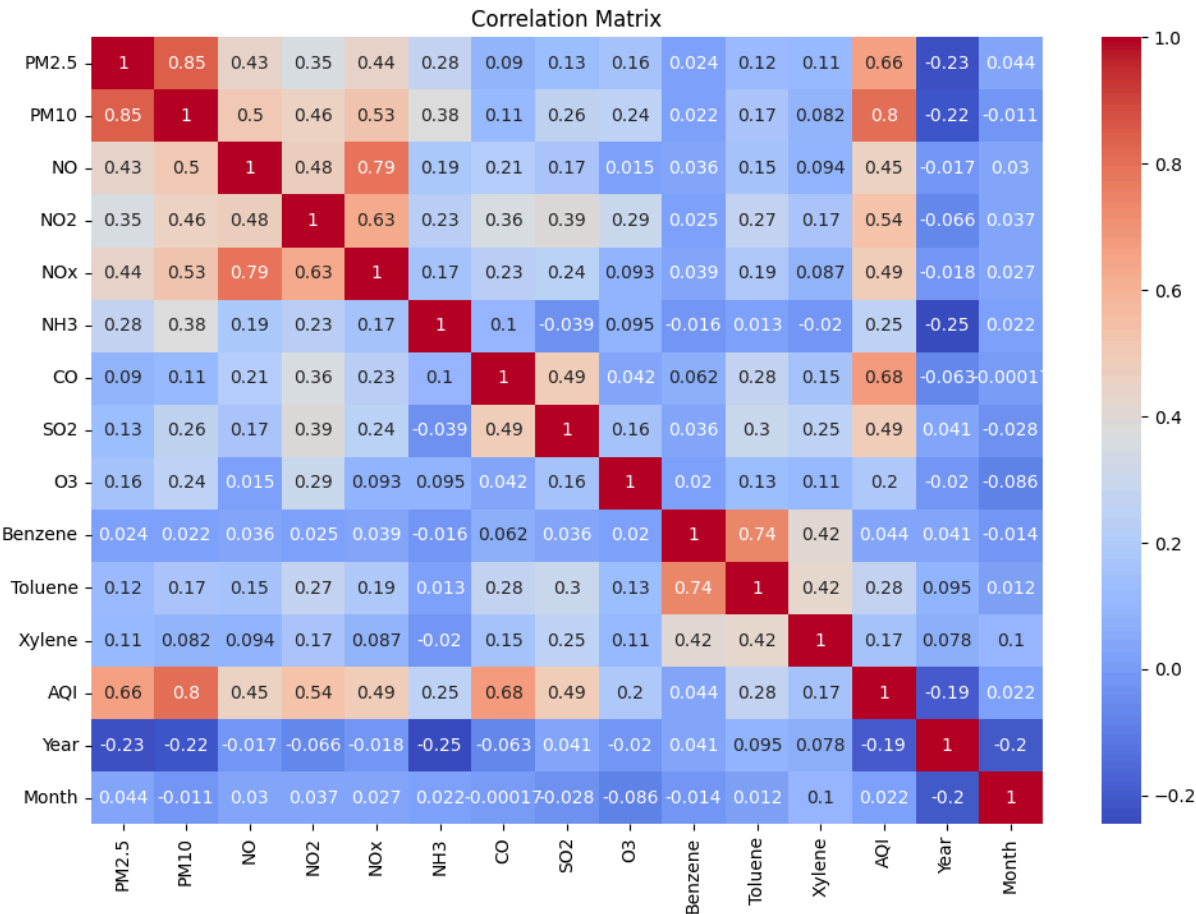
```
In [4]: numeric_cols = aqi.select_dtypes(
        include=np.number
    )

    corr = numeric_cols.corr()

    plt.figure(figsize=(12,8))

    sns.heatmap(
        corr,
        annot=True,
        cmap="coolwarm"
    )

    plt.title("Correlation Matrix")
    plt.show()
```



Lab 2 - Task 9

Briefing to the Honorable Minister

Our analysis indicates that air pollution levels vary considerably across Indian cities and exhibit clear seasonal patterns. Pollution levels tend to increase during certain months, particularly around the post-monsoon and winter periods, when atmospheric conditions trap pollutants closer to the ground.

The data also shows substantial differences between locations, suggesting that pollution is not evenly distributed across the country. A small number of extremely high readings significantly influence average pollution values, highlighting the importance of monitoring pollution hotspots separately.

Long-term trend analysis suggests that air quality has fluctuated over the years rather than improving consistently. While some periods show improvement, others indicate worsening conditions.

Recommendation: Prioritize targeted pollution-control measures in identified hotspot regions, especially during high-risk seasonal periods.

Limitation: The available datasets do not contain sufficient common geographic information to directly establish whether higher pollution causes lower crop production. Other factors

such as rainfall, irrigation, soil quality and farming practices are not included and may significantly affect crop yields.

CONCLUSION

The exploratory analysis provided valuable insights into air quality patterns across India. Trend analysis showed that pollution levels have varied over time rather than exhibiting a consistent improvement or decline. Seasonal analysis indicated that AQI levels fluctuate throughout the year, with certain months experiencing noticeably higher pollution levels, supporting the possibility of seasonal environmental influences. Correlation analysis among numerical variables revealed meaningful relationships between several pollutants, suggesting common sources or atmospheric interactions. However, due to the absence of state-level information in the AQI dataset, a direct and reliable assessment of the relationship between air pollution and agricultural production could not be established. Therefore, while the analysis highlights important pollution trends and patterns, additional integrated data would be required to conclusively determine the impact of air quality on crop yields. Overall, the study demonstrates how data analytics can support environmental decision-making while emphasizing the need to recognize dataset limitations and avoid treating correlation as proof of causation.